

ARAVIND KUMAR NALUKURTHI

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OBJECTIVE

AI Engineer and Data Engineer with 3+ years of experience building LLM powered systems, agentic workflows, RAG pipelines, and large scale ETL and data platforms using Python, Spark, SQL, and cloud native tooling. Focused on solving real world reasoning, automation, and decisioning problems.

WORK EXPERIENCE

AI Engineer Intern January 2026 – May 2026
Optimalex New York City, NY

- Built **LLM** powered **agentic AI** components using **structured prompting**, **autonomous tool calling**, and **MCP** based context routing to interpret claim data and policy rules. Improved the consistency and reliability of decision support across high volume claim scenarios by **24.6%**.
- Engineered **retrieval augmented pipelines** combining **embeddings based retrieval**, domain knowledge, and grounding checks to strengthen prediction accuracy. Reduced reasoning errors by **18%** and improved overall decision safety.
- Designed and automated **AI workflows** using orchestration tools such as **n8n**, **Dify**, and **IBM Orchestrate**, enabling reliable multi-step reasoning, routing, and human-in-the-loop review for regulated insurance workflows.
- Applied advanced **prompt engineering** and **AI workflow optimization** techniques to improve model determinism, reduce hallucinations, and ensure structured, trustworthy outputs across production-grade pipelines.

Research Assistant, Artificial Intelligence November 2024 – May 2025
Arizona State University Tempe, AZ

- Built and evaluated models for **text generation**, **summarization**, and **user intent understanding** using prompt driven development, **embeddings based retrieval**, and **model fine tuning**. Improved output relevance and model stability across multiple class modules.
- Designed **agentic workflows & retrieval augmented** prototypes that used **vector based search**, **domain knowledge**, & **evaluation loops** to interpret questions & produce context aware answers. Improved reasoning accuracy & strengthened reliability of AI assisted academic tools by 72%.

Data Engineer July 2022 – July 2024
Uber Technologies Inc (Via NINEleaps Technologies)

- Designed and operated end-to-end **data ingestion pipelines** using **Kafka**, **Spark**, and **Airflow** to capture and process **50M+ daily auction events** across Uber Mobility and Eats. Delivered clean, analytics-ready datasets powering advertiser reporting, fraud detection, and budget optimization.
- Built large-scale **ETL and ELT workflows** on **PySpark**, **SparkSQL**, **Hive**, and **Redshift**, transforming raw mobility and delivery signals into curated fact and dimension models. Enabled high-quality insights for BI and real-time decisioning systems.
- Engineered OLAP-optimized **star and snowflake schemas** with SparkSQL and HiveQL, unifying fragmented auction and campaign datasets and improving query performance by **65%**. Ensured consistent **data lineage** across Cassandra, Redis, and Hive warehouse layers.
- Implemented scalable **data quality and governance** frameworks using **AWS Glue**, metadata-driven Spark jobs, and validation rules, eliminating petabyte-scale inconsistencies and enabling reliable, self-service analytics for cross-functional teams.
- Designed and tuned high-throughput **batch and streaming** pipelines on **AWS EMR**, **S3**, and **Lambda**, optimizing partitioning, schema evolution, and storage formats to reduce pipeline runtime by **40%** and support real-time bidding, budget pacing, and fraud detection workflows.
- Integrated **CI/CD practices** using Git, automated testing, and deployment pipelines to ensure safe, repeatable releases of data workflows and Spark jobs across staging and production environments.
- Authored detailed **technical documentation**, data dictionaries, and workflow diagrams, improving onboarding speed and cross-team collaboration. Participated in **Agile ceremonies** including sprint planning, standups, and retrospectives to deliver features on predictable timelines.

PROJECTS

AI-Driven Customer Issue Triage and Resolution Agent
[GitHub](#) Python, FastAPI, LangChain, OpenAI, Pinecone, Redis, Airflow, Docker, AWS Lambda

- Built an end-to-end LLM powered triage agent that ingests support tickets, retrieves historical resolutions via **RAG** over Pinecone, and executes **agentic tool calls** (SQL lookups, policy checks, status updates) through a FastAPI action layer. Reduced manual triage load by 63% and cut first-response time from minutes to under 10 seconds across 40K+ monthly tickets.

Ads Bid Health Monitor & Auction Quality Framework
[GitHub](#) | Python, SQL, PySpark, Isolation Forest, XGBoost

- Built an auction quality monitoring system applying Isolation Forest and DBSCAN outlier detection across 500M+ daily auctions to surface a 34% eCPM variance driver and an 8x fraud signal invisible to rule-based systems, then fed those anomaly signals into an XGBoost fraud model achieving AUC 0.94 — delivering an automated bid health layer that caught fraud earlier and reduced manual investigation effort across the platform.

TECHNICAL SKILLS

Programming : Python, Java,C, C++, Scala, API Design (REST, WebSocket), Distributed Systems, Microservices, HiveQL, SparkSQL.
Data Engineering: Apache Kafka, Apache Flink, Apache Spark, Airflow, Event Streams, Job Orchestration, Data Pipelines, ETL/ELT
Databases & Storage: PostgreSQL, MySQL, Cassandra, Redis, MongoDB, Data Modeling, Schema Design, Dimensional Modeling
Machine Learning & GenAI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, LLMs, RAG Pipelines, Prompt Engineering, Intelligent Agents, Multi-Agent Workflows, Model Evaluation & Monitoring, Claude Code, Hugging Face, NLP, Deep Learning.
Cloud & Infrastructure: AWS (S3, EC2, Lambda, EMR, Glue, IAM), GCP, Docker, Kubernetes, CI/CD Pipelines, Model Deployment.
Data Platforms & Lakehouse: Databricks, Snowflake, Delta Lake, Iceberg, Unity Catalog, Data Lake Architecture, Lakehouse Patterns

EDUCATION

Arizona State University, Tempe, AZ August 2024 – May 2026
Masters in Data Science, Analytics and Engineering (Computing & Decision Analytics)

National Institute of Technology, Calicut July 2018 – June 2022
Bachelor of Technology in Electronics and Communication Engineering